

Prospects for Inertial Fusion Energy Based on a Diode-Pumped Solid-State Laser (DPSSL) Driver: Overview and Development Path

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PROSPECTS FOR INERTIAL FUSION ENERGY BASED ON A DIODE-PUMPED SOLID-STATE LASER (DPSSL) DRIVER: OVERVIEW AND DEVELOPMENT PATH

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Abstract:

It is now known with certainty that the type of fusion known as inertial fusion *will* work with sufficient energy input, so inertial fusion is really beyond the “scientific breakeven” point in many respects. The most important question that remains for inertial fusion energy (IFE) is whether this type of fusion can operate with sufficiently low input energy to make it *economically* feasible for energy production. The constraint for low input energy demands operation near the inertial fusion ignition threshold, and such operation creates enormous challenges to discover a target design that will produce sufficient energy gain. There are also multiple issues relating to the scientific feasibility of using a laboratory-type “driver” to energize a target, such as those concerning bandwidth and beam smoothing for “direct drive,” and extension of hohlraum plasma physics to the IFE scale for “indirect drive.” One driver that appears as though it *will* be able to meet the IFE requirements, assuming modest development and sufficient target gain, is a diode-pumped solid-state laser (DPSSL). We give an overview of this type of laser system, and explain what development remains for the economic production of electricity using this type of driver for IFE.

1. Introduction

With the ever-present demand for electrical power in the modern world, the technical community must prepare a development path for a technology that will be able to meet future energy demands within appropriate cost guidelines. With the ever-declining natural resources of fossil fuels and the increasing environmental impact of burning fossil fuels, the technologies of interest are primarily those having an essentially inexhaustible and environmentally tolerable fuel source—breeder fission reactors, solar power, or fusion. The key questions for any technology development relate to whether the technology provides a practical solution. Such practicality involves acceptable technological risk and sufficiently low environmental impact.

In this paper, we consider only the fusion technologies. Of these, inertial fusion has already proved itself to be viable, but for input energies much above the amount that would permit economic operation. Although this fact is not well known, the scientific feasibility at such energies has already been demonstrated in underground nuclear tests in Nevada. What remains is to demonstrate that inertial fusion can operate for much smaller input energies, where economic feasibility can be achieved, but where scientific feasibility issues arise in connection with using a laboratory driver.

Because of relatively recent research, an advanced type of driver has emerged that appears to be capable of providing inertial fusion with a practical driver solution. This driver is a diode-pumped solid-state laser, i.e., a DPSSL (pronounced “dip-sol”). In this paper, we explain what inertial fusion is, and provide an overview and development path for a power plant driven by a DPSSL. Such a development path shows that economic feasibility for IFE depends primarily on the achievement of high target gain.

2. What is fusion?

Fusion is the nuclear process in which very light nuclei like hydrogen are combined (“fused”) to make a slightly heavier nucleus such as helium. Fission, in contrast, is the nuclear process in which very heavy nuclei like uranium are broken up through radioactivity. Both fusion and fission release energy through the well-known Einstein equation $E = mc^2$, because the combined mass of the products is less than the sum of the masses of the starting nuclei—hence, mass has been converted into energy. One difference between fusion and fission is that fusion releases about four times more energy per unit mass, so fusion is preferred from an energetics standpoint. Fusion also involves less radioactivity because, unlike fission, the process itself is not inherently radioactive, and any induced radioactivity (e.g., from neutron emissions activating nearby structures) can be kept to a minimum by the use of non-activating materials. Fusion is also not subject to explosion (as at Chernobyl). Fusion is also not subject to the “China syndrome” (where the uranium fission fuel can melt down through its containment if emergency cooling fails); no China syndrome is possible for fusion because only light elements are used, so the fuel heat content is very low. Fusion thus offers many advantages over fission.

The drawback for fusion has been the difficulty in developing the technology. In order for fusion to occur, the temperature of the constituents must be raised to about 100 million degrees K. At this temperature, the fusion material cannot be contained by ordinary mechanical vessels (which would melt), but must be confined either magnetically or inertially. The length of time that the fuel must be confined depends on the density to which the constituents are compressed, because the reaction rate is proportional to the product of the confinement time and the square of the density. Magnetic confinement involves a low-density gaseous fuel, so it needs seconds or more of confinement; inertial confinement involves densities nearly a thousand times liquid hydrogen densities, so it requires confinement times of 10^{-9} second or less. The different types of fusion are therefore designated by the type of confinement. Fusion occurs in the center of the sun, where gravitation provides the confinement and the high temperatures.

3. What is inertial fusion?

For inertial fusion, a capsule is prepared that is essentially a spherical “fuel” shell of hydrogen isotopes surrounded by what is called an ablator (i.e., a layer that can be burned away). Such a capsule is irradiated with beams of energy from a “driver.” If the irradiation is directly on the capsule, the type of drive is said to be “direct drive.” If the driver beams are focused into a small metallic cylinder (a hohlraum) that heats up and irradiates the capsule with x rays, the type of drive is said to be “indirect drive,” and the hohlraum with the capsule mounted inside are said to be the “target.”

The driver beams (or the x rays) heat up the surface of the ablator to such an enormous temperature that the surface material blows off radially at great speed. Because for every (momentum) action there is a (momentum) reaction, the blow off causes an inward compression (implosion) of the capsule. As the capsule implodes, the density and temperature at its center increase to the point at which the fusion reactions

begin (i.e., the fuel “ignites”). At this time, the inward-moving fuel is confined by its own inertia, because the inward-moving fuel tends to continue moving inward. This type of fusion thereby goes by the name inertial confinement fusion (ICF).

As the fusion fuel “burns,” the nuclei fuse together and release energy in the form of fast (14-MeV) neutrons, x rays, and particulate kinetic energy (the energy of the moving products). This sudden release of energy stops the implosion, and reverses it to form a miniature explosion.

For a power-plant application, an ICF capsule is about the size of a marble, and its energy release is about 1/4 ton of TNT equivalent. Such explosion sequences are repeated at the rate of about 5 to 10 targets every second. The energy from such explosions is collected as heat in the wall of a spherical fusion chamber that contains the explosions. This heat can be used to operate a power plant with a steam turbine in much the same way as a coal-fired plant. A fusion plant is hence very similar to a regular power plant, except that a fusion plant needs a driver to generate beams of energy to compress the fusion fuel to the required conditions, and a fusion chamber.

It is noteworthy that inertial fusion will work with different fuels, although the type easiest to ignite consists of equal amounts of two isotopes of hydrogen: deuterium (D), whose nuclei contain one proton and one neutron, and tritium (T), whose nuclei contain one proton and two neutrons. Although tritium is somewhat radioactive, and hence DT fuel is somewhat radioactive, the next generation of inertial fusion could use just deuterium (DD). This type of fuel is not radioactive, and can be extracted from sea water at the rate of about 1/15 gram per gallon, which has the releasable energy content of nearly 2 tons of TNT equivalent (i.e., about 8 capsules worth). The vastness of the oceans therefore provides an essentially inexhaustible supply of fusion fuel. The disadvantage of DD fuel is that it is harder to ignite, requiring about 3 times the ignition temperature (and hence a larger power plant to gain economy of scale and preserve economic feasibility, and a different target design).

4. The Advantages of ICF

At this time, ICF has many advantages over magnetically confined fusion (MCF). First, ICF is now known to work with sufficient energy input. Underground experimental tests in Nevada have allowed the demonstration of excellent performance, putting to rest the fundamental questions about the basic feasibility of achieving significantly more energy output than input (i.e., of achieving high gain). Basic scientific feasibility is therefore no longer an issue for ICF, with sufficient energy input.

Second, DT fusion produces fast neutrons whose energy content can be collected—and the environment shielded from induced radioactivity—only by the introduction of shielding. For ICF, this shielding is minimal because the source of the neutrons is essentially a point (the center of the imploded capsule). The shielding can therefore be placed around the exterior wall of the fusion chamber as a spherical shell, and can even be designed to serve both as shielding and as (or in conjunction with) an absorber of neutron energy for the transfer of heat energy to a heat exchanger. For MCF, however, the source of the neutrons is a large extended source (e.g., the entire doughnut-shaped interior cavity of a tokamak). Consequently, the geometry of the shielding for MCF cannot be a spherical shell, so considerably more shielding is

required. Third, because of the low-density nature of the MCF fuel, and the large amount of shielding required, a power plant utilizing MCF is much larger and hence much more expensive than we think a power plant utilizing ICF will be.

Fourth, ICF has greater physical separability between the location of the fusion reactions and the emplacement of the high-technology equipment that energizes the fusion fuel. Fifth, ICF offers a clearer approach to maintenance—continuous beam maintenance may even be feasible by including extra driver beams. Sixth, ICF offers more control, because one driver can feed multiple fusion chambers, and variable repetition rate can be employed. Seventh, ICF has an attractive development path.

5. Why don't we have an ICF power plant now?

Before utility companies will become interested in inertial fusion energy (IFE)—that is, to make energy in a power plant based on ICF—we must demonstrate that IFE can be *economically* feasible. The main problem for IFE (and fusion in general), other than the complex nature of the technology itself, is the great difficulty in designing a power plant that has a low enough construction cost to permit economic feasibility. The reason for this is that an IFE plant includes extra equipment that a fossil-fuel plant does not have. An IFE plant has a driver and a fusion chamber *in addition to* the thermal (steam-turbine balance-of-plant) equipment that is common to IFE as well as to other types of power plants. Because the fuel costs for IFE should be almost negligible, the combined cost of the driver and the fusion chamber for IFE must be compared with the cost of the boiler in a fossil-fuel plant and the operational cost of securing fossil fuel. Such a comparison indicates that IFE must demonstrate economic feasibility by designing a driver and a fusion chamber that are low enough in cost.

The size, and therefore the cost, of a fusion chamber is dependent primarily on the plant's thermal power, and only weakly dependent on the energy that the driver beams must focus on the target. In contrast, the cost of the driver is highly dependent on the delivered energy. Thus, for a given size of power plant, the economic competitiveness of IFE depends on whether the technology will function reliably at a driver energy that is low enough to permit the cost of the driver to be minimal. The amount of driver energy required is in turn determined by the target gain (i.e., the ratio of the energy produced by the target and the energy delivered to the target by the driver beams). One key to the economic feasibility of IFE is therefore the ability to operate ICF as close to its ignition threshold as is possible and still maintain reliability. Such operation introduces technology constraints on finding the target design that will produce a high-enough energy gain (>100) to allow a small-enough driver (1 to 2 megajoules).

The lack of sufficient target gain has made another issue become a key to the economic feasibility of IFE. This secondary key is the scientific feasibility of using a laboratory driver to energize an available type of target. Targets requiring more energy require a larger driver, which is more difficult to develop because of the technological constraints in generating more energy in temporally shaped spatially smooth beams and in delivering this energy efficiently to a target. Because the most reliable target designs deliver calculated gains significantly less than 100 for small driver energies, a larger (~ 2 megajoule) driver had to be developed. It is the difficulties with the technological

challenges dealing with target gain and the feasibility of the required driver that have constrained the early development of IFE.

6. Different ICF driver options and issues

The pursuit to find an inexpensive driver has focused primarily on a heavy-ion accelerator that will deliver beams of fast-moving heavy ions (e.g., lead ions). The interest in such machines arises from their high efficiency (about 35%) when used for high-energy physics research, because an efficiency of $\geq 10\%$ is thought to be needed for economic feasibility for IFE. Considerable research effort has therefore been directed to try to discover by how much (if any) the efficiency will be degraded in generating, transporting, and focusing a high-current heavy-ion beam on a target. So far, although positive results are expected, it is not known whether the potential problems associated with the accelerator plasma effects and target space-charge effects will be resolved. Uncertainty arises primarily because of the lack of development—no machines have been built that deliver the applicable beam intensities, energies, and pulse widths, and no one has directed such driver beams onto a target. There are also research efforts for KrF lasers and light-ion accelerators, but these efforts today suffer from uncertainties arising from technologies that are even less developed. KrF systems have many development problems (optics damage, etc.) associated with producing a high-power beam on target, and with the short lifetime of some of its components. Light-ion technologies are only in their infancy. Thus, these other technologies may eventually prove fruitful, and progress is being made, but it is too early to know *with certainty* whether conceptual solutions to all of the problems will be found.

Research conducted at the Lawrence Livermore National Laboratory (LLNL) over the past several years (see below) enabled us to conduct a full systems analysis of an IFE power plant based on a different type of driver—a diode-pumped solid-state laser (DPSSL).⁽¹⁾ This study includes the essential concepts for irradiating a target with driver beams, and is hence a comprehensive systems study. Such a study was possible because of the significant commonality that DPSSLs have with the flashlamp-pumped solid-state lasers (SSLs) currently being employed in ICF research. We expect that added understanding will come through research on the National Ignition Facility (NIF) now under construction at LLNL. We are therefore concerned in this paper with a DPSSL driver.

Nevertheless, our systems study identified many issues that must be addressed for DPSSLs (particularly target gain—see below), so research at LLNL is being pursued on both heavy ion beams coupled to indirect-drive targets and DPSSLs for either direct or indirect-drive targets. This dual track approach is justified by the present uncertainties in (1) the target performance that will eventually be achievable, and (2) the driver technology development. Assuming that suitable high-gain targets can be developed, we believe that our study indicates that the DPSSL technology will be suitable for the IFE mission.

Recall that the word “laser” is an acronym meaning Light Amplification by Stimulated Emission of Radiation. A laser thus works by having a “gain medium” with a special dopant that can be energized up to an excited state by a light source such as a flashlamp (which is similar in concept to the strobe light used to time automobile engines). The gain medium can then be stimulated to release its energy by the passage of

a light pulse whose photon energy matches the energy difference between the excited state and the ground state in the gain medium. Such stimulation adds energy to the light pulse, and thus provides amplification.

Our DPSSL driver is similar to the flashlamp-pumped lasers currently being operated at the University of Rochester (the Omega laser) and at LLNL (the Nova laser), where laser beams have imploded capsules. The only significant exceptions are that our DPSSL uses crystals as the gain medium instead of the usual neodymium-doped glass (Nd:glass), and it uses light-emitting laser diodes instead of flashlamps to pump the gain medium. These substitutions are needed for the rep-ratable operation required for an IFE driver.

The importance of the similarity of a DPSSL to an ordinary SSL is far-reaching. Because a DPSSL is related to existing SSLs in terms of its basic architecture and optical components (other than the gain medium and pumping optics), the experimental success for a DPSSL can draw from the significant experimental base already established with existing SSLs. This means that, in essence, target physics and laser operation have already been investigated to a considerable degree for a DPSSL.

7. Advances made since the 1st Symposium in November 1994

Significant advances have been made over the past several years in both laser design and target-physics design. The research advances at LLNL that have permitted the DPSSL to achieve credence as a potential IFE driver are the following.

1. The development of a novel laser gain medium with better thermo-mechanical properties, Yb³⁺-doped Sr₅(PO₄)₃F [called Yb:S-FAP], whose ≥1.10-ms storage lifetime is at least four times longer than the 0.3-ms lifetime for Nd:glass, thereby reducing the diode pump array expenditure by roughly a factor of 4.
2. The manufacture of efficient laser diode pump sources to pump the new gain medium up to its excited state, so that laser operation at efficiencies of ≥10% are possible (instead of the ≤1% attainable now with Nd:glass). Such laser diode manufacturing techniques make it feasible to eventually scale to acceptably low manufacturing costs at power-plant-level production volumes.
3. The conceptualization of fused-silica final laser optics heated to temperatures of 400°C or more to continuously anneal out the structural defects and optical discoloration induced by neutron interactions, thereby circumventing the low-lifetime constraint on the use of these optics in environments with high neutron fluences—which would otherwise quickly turn the optics opaque. (This solution works for a DPSSL, but not for KrF wavelengths.)
4. The development of a large-aperture gas-cooled Pockels cell that permits the realization of a compact multipass amplifier architecture by serving as an optical switch to switch a beam out of a multi-pass amplifier cavity.
5. The calculational and experimental validation of gas-cooling techniques to cool the gain medium crystals, thereby showing that the thermally induced optical distortions in the gain medium can be reduced to a simple beam-steering effect, while removing the waste heat.
6. The experimental operation at LLNL of a 2 joule DPSSL prototype with a single 2×2×0.5-cm Yb:S-FAP crystal with helium gas cooling operated with a 1-ms diode

pump pulse in a free-running long-pulse (1-ms) mode that verified the basic thermal operation of this type of DPSSL. These experiments⁽²⁾ demonstrated an intrinsic laser slope efficiency of 67% (theoretical limit is 86%), 25-Hz operation (more than twice that needed for IFE), and facial thermal fluxes several times that needed for 10-Hz IFE operation.

These research results provide potential solutions to all of the issues previously thought to preclude the use of a solid-state laser driver for IFE. The Mercury Project at LLNL, to be described below, will demonstrate the short-pulse extraction and rep-ratability of a DPSSL at 100 joules output. Thus, a DPSSL is well on its way to the actual experimental demonstration of the capabilities required for an IFE driver. Nevertheless, the price of laser diodes to pump the DPSSL gain medium is currently a factor of about 100 too high to permit economic feasibility at the target gain expected for an IFE target. Consequently, there is a need to promote the manufacture of lower-price diodes, and a need to encourage the development of higher-gain targets. On the other hand, we believe that the semiconductor industry will be able to adequately respond to the IFE cost goals for laser diodes, given the existence of a large-enough market, because the price of diodes is falling in the same way that was observed for the cost of semiconductor integrated circuits. As a result, most of the burden for economic feasibility of IFE may fall on the need for higher-gain targets.

The research advances over the past several years that have permitted enhanced credence in our ability to design ICF targets are the following.

1. Experimental demonstration, primarily at LLNL's Nova laser, of our ability to control laser-plasma instabilities (e.g., stimulated Raman scattering and stimulated Brillouin scattering), hohlraum energetics, capsule symmetry, and capsule hydrodynamic instabilities for targets that were operated at tens of kilojoules of blue laser light, but considered to be hydrodynamic equivalents to those at IFE scale.
2. Enhanced experimental validation of Lagrangian and quasi-Eulerian modeling codes used to simulate laser-hohlraum drive, symmetry, and capsule implosions, thereby improving the credence in the modeling predictions for targets at NIF and IFE scales.
3. Demonstrated understanding on Nova of implosions of capsules having high NIF-like convergence ratios (i.e., capsules that implode to a small fraction of their initial radius).
4. Demonstrated understanding on Nova of hydrodynamic (Rayleigh-Taylor) instabilities in quantitative agreement with numerical modeling for planar foils and deliberately perturbed 3D capsule surfaces.
5. The theoretical introduction of a "fast ignitor" target that may be capable of gains well above 100. Such a target employs standard capsule compression, but then the use of a 100-picosecond 10^{18} W/cm² laser pulse to "channel" through to the fuel core, and a 5-picosecond 10^{20} W/cm² laser pulse to ignite the fuel through the channel.

Although much progress has been made, more development is of course required.

8. Near-term development—the Mercury Project

Extension of the current research base to an IFE application requires development in both target physics and DPSSL technologies. Although our IFE design for a DPSSL [see ref. 1] is based rather extensively on the mature experimental technologies for solid-state lasers, the data do not always pertain to the realm of interest for IFE. The same is true for targets. The key issues remaining at this time therefore relate to the extension of this experimental base to the relevant realm, and for a DPSSL, to the particular use of the Yb:S-FAP gain medium. These issues can be addressed and the risk for a DPSSL driver can be reduced through a development path that begins with small-scale experiments in the near term, and then expands greatly with the startup of the National Ignition Facility (NIF).

For the near term, Mercury is a three to four-year project at LLNL to design (FY97), build (FY98–FY99), and operate (FY99–FY00) a 100-joule 10-Hz 1-ns DPSSL that will demonstrate efficient rep-rated short-pulse extraction. Mercury will also demonstrate compatibility with scaling to >1 kilojoule in future single-beam systems, and eventually to IFE systems at the megajoule scale. The goals include an overall electrical efficiency of $\geq 10\%$, a final focus less than 5 times the size of a diffraction-limited beam, and pulse lengths of 1 to 10 ns. What this means is that the Mercury laser will demonstrate the utility of a DPSSL as an IFE power plant driver in many respects, but not in total energy output—it will demonstrate the relevant pulse duration, the needed laser efficiency, and the required thermal management at a repetition rate of 10 times every second. It will also address the issues relating to the fabrication of large diode arrays.

It is important to note that the efficiency of interest for an IFE driver is the total efficiency in driving a target, which should be $\geq 10\%$ (actually, it is the product of the efficiency and the target gain that should be roughly ≥ 10). This means that the efficiency must include the efficiency factor for the coupling of the beam energy to the fusion target. Such a factor, which depends on the quality and lateral spread of the beams at the target, is about 90% for a DPSSL, based on experiments at Nova and at other laser facilities.

9. Intermediate-term development: The Role of the National Ignition Facility (NIF)

The NIF is a single-shot 192-beam facility under construction at LLNL to demonstrate that a driver with ≤ 1.8 megajoules can ignite a capsule and provide moderate target gains of 1 to 10. (The NIF also has missions relating to stockpile stewardship for thermonuclear weapons.) This facility is expected to come on line in the year 2002, and continue operation for one or two decades. Although the NIF will be implemented with Nd:glass lasers, the results for both the target physics and laser multi-pass architecture will apply directly for a DPSSL-driven IFE plant.

The NIF is expected to be a pivotal facility for ICF research, because one of its missions is to demonstrate ignition and propagating burn in the laboratory. Its most significant accomplishment relating to IFE is expected to be the determination of the exact ICF ignition threshold for each major type of ICF target, and hence the operating

point for reliable IFE operation. Next in importance for IFE should be the insight relevant for the design of an IFE target with a gain ≥ 100 . Research on the NIF is expected to provide the needed understanding for such things as the precise requirements for shaping each laser pulse in time, how spatially smooth each laser beam must be (i.e., the required bandwidth), whether direct or indirect drive is better, and complete understanding of the relevant plasma instabilities inherent in the coupling of the laser energy into a hohlraum and generating x rays to implode a capsule in a stable manner.

To be certain that DPSSL gain-medium crystals can be grown to large enough sizes for IFE applications, and to test the operation and thermal management of rep-ratable DPSSLs at the kilojoule scale needed for IFE, a sequence of single-beam intermediate-term DPSSL prototype facilities is being considered as one option: Venus offering about one kilojoule, Terra offering 10 to 30 kilojoules, and (eventually) Helios being a rep-rated producer of net electrical power. These facilities *could* be operated in parallel with the operation of the NIF, whose single-beam energy is about 10 kilojoules.

Note that only single-beam facilities are required for most DPSSL development activities because a DPSSL system is entirely modular. DPSSL functionality can therefore be validated at the component level or at the single-beam level, thus permitting experimental verification at reduced development cost. In addition, many of the novel laser-diode technologies envisioned for an IFE driver are relevant to a variety of scientific, industrial, and military applications and will therefore continue to advance in parallel with ICF developmental efforts.

10. Future development path: What are the remaining issues?

After the NIF and the DPSSL developmental facilities have been operated, the fusion community needs to construct a complete engineering demonstration facility (EDF) to test the engineering issues for IFE, and to prove the effectiveness of fabricating and delivering targets at the rate of roughly 10 targets every second to be tracked and then illuminated with DPSSL beams at the center of the fusion chamber. There are also issues connected with the lifetime of the first wall of the fusion chamber that could involve significant engineering problems, as well as issues pertaining to the clearing of plasma debris from one explosion prior to the next shot 100 ms later. There are also significant issues relating to the final optics that would be exposed to neutron (as well as x-ray and plasma-debris) emissions.

One particularly important issue concerns whether DPSSL optics can withstand the 10^{10} pulses expected during the 30-year lifetime of an IFE plant, even if the potential problems regarding neutron and x-ray damage can be completely solved. Even though a DPSSL operates at fluences only a factor of about 4 higher than off-the-shelf lasers capable of $>10^8$ pulses, more data are needed to convince us that a DPSSL can operate without damage at the IFE conditions identified in our systems study (e.g., a peak fluence of $\sim 20 \text{ J/cm}^2$ at $\sim 9 \text{ ns}$ effective square pulse duration at a wavelength of 349 nm for the final optic).

There are also issues relating to the design of a fusion chamber that can accommodate a laser driver. Laser beams are usually expected to enter a fusion chamber at large cone angles, instead of in two opposing small-angle clusters, but such a beam configuration is not thought to be compatible with liquid first wall designs that permit a

high plant availability fraction. This issue, however, is directly associated with the amount of target gain—if sufficient target gain is achieved, narrow cone angles can be used and the issue becomes inconsequential. This is one more argument that emphasizes the need for higher-gain target designs. Other issues that must be resolved concern target injection (and the associated beam-alignment tolerances), and pulsed-neutron damage to materials.

11. Conclusions

The interest in a DPSSL as an IFE driver is growing steadily. ICF is now *known* to work at sufficient energy input. Scientific feasibility for IFE would therefore seem to be assured in many respects, with the main issue being primarily *economic feasibility*. Unfortunately, all of the known designs for reliable targets require so much driver energy that such economic feasibility has become an issue of scientific feasibility both for the targets and for the ability of the driver to supply the appropriate beams to the targets. In particular, the economic demand for small driver energies forces IFE to operate just above the ignition threshold, whereas target gain there is insufficient even for driver energies above one megajoule. The larger driver energy introduces technological constraints in generating more energy in temporally shaped spatially smooth beams and in delivering this energy efficiently to a target. Great effort must therefore be expended to improve target gain for near-threshold implosions and to understand these technological constraints of a laboratory driver. The NIF is meant to provide most of the answers to the remaining questions relating to this current challenge for laboratory IFE research efforts.

Meanwhile, the DPSSL has emerged as a candidate driver that appears as though it *will* be able to meet the IFE mission requirements, given modest future development and suitable target gains. With the research advances over the past several years at LLNL, we now know that a DPSSL need not be subject to the major issues thought to restrict the use of a solid-state laser as an IFE driver. The main inhibiting factor for the use of a DPSSL today is the high price of the laser pump diodes, but that is expected to drop as the market for laser diodes expands. Consequently, the only known potential problems for the use of a DPSSL as an IFE driver are (1) the development of targets that have sufficient gain to offset both a target operation near ICF ignition threshold and the reduced-cone-angle beam illumination suitable for wetted-wall fusion chambers, and (2) the development of good solutions for final optics protection. In other respects, DPSSL technology relies on the mature solid-state laser experimental base derived from two decades of experience. DPSSL technology is also a natural extension of the NIF towards IFE applications. There remains, of course, the demonstration that DPSSL technology will work at the IFE scale, but the related questions are not at the conceptual level. In addition, the Mercury Project is poised to demonstrate the rep-ratable short-pulse-extraction characteristics of the Yb:S-FAP gain medium for a 100-joule output, as well as gas-cooling and fabrication of large laser diode arrays. It would therefore appear that DPSSLs are indeed a viable candidate for an IFE driver.

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